Deadman Power Protector

Hardware Bible

Hardware revision A: parts, circuits, wiring, cables and safety

Release	Version 1 (tag `v.1`), declared 2026-11-12
Hardware revision	A
Firmware version	`v.1.0.20261113T000224Z`
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Author	Zeroshin

The Deadman Power Protector cuts a workshop power bar when nobody is at the bench. A Shelly 1PM Gen4 relay switches the bar. A bench node with a radar presence sensor decides, and renews a lease with the Shelly; if the renewals stop, the Shelly cuts the bar by itself. It is for a maker who leaves soldering irons and supplies on and wants them off when the bench is empty, without being interrupted while working.

Colophon

Item	Value
Release	Version 1 (tag `v.1`), declared 2026-11-12
Exported	2026-11-12
Hardware revision	A
Firmware version	`v.1.0.20261113T000224Z`
Author	Zeroshin
Licences	This document: `CC-BY-SA-4.0`. Hardware design (KiCad files): `CERN-OHL-S-2.0`. Firmware and the Shelly script: `GPL-3.0-or-later`.
Root repository commit	`115caae`

Callouts in this document:

> ⚠ DANGER. A hazard that will kill or seriously injure, and how to avoid it.

> ⚠ WARNING. A hazard that could kill or seriously injure, and how to avoid it.

> NOTICE. Risk of damage to the device or to what it is plugged into, and how to avoid it.

> Note. A fact not to miss. It is not a hazard.

The other book, the Markdown editions of both books and every source file are in the Deadman Power Protector repository: start with its `README.md`.

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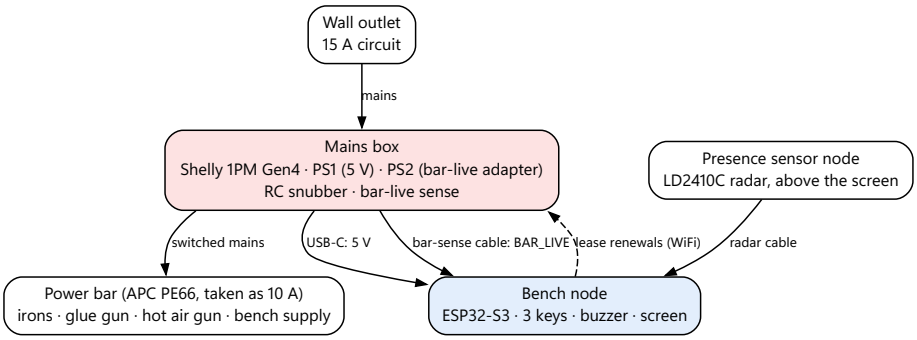
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The Deadman Power Protector is a cutoff for a workshop power bar. It turns the bar off when the person at the bench leaves, not on a clock. A Shelly 1PM Gen4 smart relay in a mains box switches the bar; an ESP32-S3 bench node with three keys, a buzzer and a small screen decides when to cut, warns before every automatic cut, and keeps renewing the Shelly's permission (the lease) to stay on; an LD2410C radar in a presence sensor node tells the bench node someone is there. If the bench node stops renewing the lease, the Shelly turns the bar off by its own timer. This file describes the hardware of hardware revision A: the parts, the circuits, every connection, the cables, the limits and how to change it safely.

1 What it is

Figure 1. The three units and what joins them.



The Deadman Power Protector has three units:

Table 1. What it is.

Unit	Where	What it does
Mains box	Between the wall outlet and the power bar	Switches the bar with U1, the Shelly 1PM Gen4 smart relay. Feeds the bench node. Senses, on its own, whether the bar is live
Bench node	At the bench	An ESP32-S3 devkit (U5) with three keys, a buzzer and a small screen. It decides when to cut, warns, and keeps renewing the Shelly's lease
Presence sensor node	Aimed at the bench and its chair	An LD2410C radar (U4) that tells the bench node someone is there

- The Shelly keeps the bar on only while the bench node keeps renewing a lease. If the bench node dies, is unplugged or loses the network, the renewals stop and the Shelly turns the bar off by its own timer. How the lease works is in the Firmware Gospel, the Firmware Gospel.

- Two cables join the mains box to the bench node: a USB-C cable carrying `5VDC` and `GND`, and a 3-pin cable carrying the bar-live signal `BAR_LIVE`. One 5-pin cable joins the bench node to the presence sensor node.
- The bar-live sense is a second, independent signal that the bar is live, read from the Shelly's output through an optocoupler, so the Shelly is not believed entirely.

2 Safety

2.1 Mains is inside the mains box

> ⚠ DANGER. The Deadman Power Protector's mains box carries 120 V mains whenever its plug is in the wall. PS1, the bench node supply, and the Shelly's own supply are on the unswitched line, so turning the bar off does not make the box safe. Unplug the box before opening it.

- In this build the mains box enclosure is a 3D-printed box, not a listed electrical box. Mains and low-voltage wiring share it, kept at least 3 cm apart, and no wire inside can move.
- No touchable wire may be at mains potential. The Shelly's SW input is referenced to mains. Never wire a key, or anything outside the mains box, to it. In the Deadman Power Protector, SW is not connected.
- The only path from the mains box to the bench node is the low-voltage side: the USB-C `5VDC` and the output of U2, the PC817 optocoupler. Nothing at mains potential leaves the box.
- Changes to a building's fixed wiring are outside this build and need a qualified electrician.

2.2 The switching device has no Canadian mark

U1, the Shelly 1PM Gen4 used here, is the European model, `S4SW-001P16EU`. It carries no Canadian safety mark (no cUL, cULus, cETLus or CSA). The build uses it anyway; that carries electrical-code and insurance risk.

2.3 What a cut does and does not do

- The Shelly's relay is non-latching: it is open whenever its coil is not driven. After a power loss the Shelly comes back with the bar off.
- There is no second relay in series. The Shelly's relay is the only break in the line: the Shelly is judged reliable enough on its own. If the Shelly's firmware hangs with its relay on, or its contacts weld shut, the bar cannot be cut; the bench node sounds an alarm instead (the alarms are described in the Firmware Gospel).
- Cutting power does not cool anything. An iron stays hot for a while after the cut. An iron stand and a smoke or heat alarm in the shop still matter.
- There is no OFF key. To stop everything at once, unplug what is in use from the bar. A press of ON held 1.5 s cuts the bar (described in the Firmware Gospel, the Firmware Gospel, chapter "The keys and LEDs").

2.4 Sizing and protection

- Every mains part is sized to carry 15 A continuously.
- No fuse is drawn or fitted in the mains box's own wiring (whether the Shelly has one inside: chapter "The switching device (U1, Shelly 1PM Gen4)", "Inside the Shelly"). The Shelly switches off above its current limit, 15 A, which is a setting of the Shelly (described in the Firmware Gospel).
- The power bar downstream is the weakest part, taken as 10 A (see chapter "Power, loads and sizing").

2.5 What must never change in the mains box

1. Every mains part and wire is sized for a full 15 A continuously: 14 AWG, and nothing that lowers that limit.
2. PS1, the bench node supply, is fed before the Shelly; PS2, the bar-sense adapter, after it. Swapped, the bench node dies at every cut, or `BAR_LIVE` reads live while the bar is off.
3. The snubber (R1 and C1) stays from the Shelly's output O to neutral, across the load, and never across the relay contacts: there it would leak about 4.5 mA into the bar while the bar is off.
4. Nothing is wired to the Shelly's SW terminal. It is at mains potential.
5. Earth runs unbroken from the plug P1 to the receptacle J1.
6. Mains and low-voltage wiring stay at least 3 cm apart, and no wire inside the box can move.
7. The only path from the box to the bench node is the low-voltage side: the USB-C `5VDC` and the optocoupler's output. Nothing at mains potential leaves the box.

Every connection and every part value stays as drawn in the schematic, except as chapter "Substitutes and safe changes" allows.

3 Power, loads and sizing

- Wall circuit: the Deadman Power Protector is built for a 15 A, 120 V, 60 Hz wall circuit.
- Sizing rule: every mains part is sized to carry 15 A continuously, not just the expected load, so the device is never what limits the circuit. No part chosen for the mains box may lower the 15 A of the box itself.
- The power bar plugs into the device's receptacle J1; it is not part of the device. The bar the design was sized against is an APC PE66, taken as 10 A (retailer listings give "1200 Watts at 120V", which is 10 A). So the bar, not the device, sets the limit at 10 A, and the bar is accepted as the weak link. The expected load stays under it.
- Loads: what is plugged into the bar is not part of the device, and its ratings do not matter to it beyond the limits in this section. Loads with a transformer, such as a linear bench power supply, are inductive; they are

the reason for the snubber (see chapter “The switching device (U1, Shelly 1PM Gen4)”).
- The Shelly’s own current limit, 15 A, is a setting (described in the Firmware Gospel).

4 Parts

4.1 Mains box

Table 2. Mains box.

Reference	Function	Part, value and rating	Polarity or orientation
P1	Mains plug and cord	NEMA 5-15P plug on 14/3 SJT cord	L, N, ground
U1	Switching device	Shelly 1PM Gen4, model `S4SW-001P16EU` (European model); relay 16 A at 240 VAC	Pins 1 to 7, left to right: O, SW, L, L, L, N, N
J1	Bar receptacle	NEMA 5-15R	L, N, ground
P3	Mains box USB-C socket, in the box wall	USB-C, VBUS and GND used; D+ and D− not connected	—
R2	Optocoupler LED resistor	1 kΩ	None
D1	Reverse diode across U2′s LED	1N4148	Reversed across U2′s LED: cathode on U2′s anode
R3	`BAR_LIVE` pull-up to `S3_3V3`	10 kΩ	None
C4	`BAR_LIVE` filter capacitor to ground	100 nF, 10 V or more	None
U11	Mains box bar-sense header	JST-XH, 3 pins	See chapter “Cables”
D3	Bar-live LED	Green, 5 mm	Anode to R12, cathode on `SENSE_GND`
R12	Bar-live LED resistor	1 kΩ	None
D4	Reverse diode across D3	1N4148	Reversed across D3: cathode on D3′s anode
—	Mains joints	Wago connectors (believed to be model 221)	—

4.2 Bench node

Table 3. Bench node.

Reference	Function	Part, value and rating	Polarity or orientation
U5	ESP32-S3 devkit	YD-ESP32-S3, N16R8 (16 MB flash, 8 MB octal PSRAM), USB-C: “ESP32-S3-DevKitC-1 N16R8, USB-C (YD-ESP32-23 2022-V1.3)”. Its IN-OUT jumper is open	Wired by its printed pin names
P2	Bench node USB-C socket	USB-C, wired to U5′s native USB-C port by a short pigtail	—
C2	Bulk capacitor on `5VDC`	100 µF, 25 V, polarized	Pin 1 positive, on `5VDC`
C3	Decoupling capacitor on `5VDC`	100 nF, 10 V or more	None
C7	Bulk capacitor on `S3_3V3`, at U5′s 3V3 pin	10 µF, 10 V or more, polarized	Pin 1 positive, on `S3_3V3`
C8	Decoupling capacitor on `S3_3V3`, at U5′s 3V3 pin	100 nF, 10 V or more	None
U3	Screen	0.96″ OLED module, 128 × 64, SSD1306 driver, I²C; as built printed “GM009605 v4.2”, maker unknown	Wire by its printed pin names (VCC, GND, SCL, SDA); boards differ in pin order
SW1	ON key	Momentary switch with a yellow 3 mm LED; as built, Cherry MX type	Pins 1 and 2 the switch contact; 4 LED anode; 3 LED cathode
SW2	OVERRIDE key	As SW1	As SW1
SW3	FN key	As SW1	As SW1
R4	ON key LED resistor	1 kΩ	None
R5	OVERRIDE key LED resistor	1 kΩ	None
R6	FN key LED resistor	1 kΩ	None
U6	Radar cable header	JST-XH, 5 pins	See chapter “Cables”
U7	Buzzer cable header	JST-XH, 3 pins	See chapter “Cables”
U8	Bar-sense cable header	JST-XH, 3 pins	See chapter “Cables”
U9	Screen cable header	JST-XH, 4 pins	See chapter “Cables”
U10	Keys cable header	JST-XH, 7 pins	See chapter “Cables”
U15	Screen-end connector of the screen cable	JST-XH, 4 pins	See chapter “Cables”

4.3 Buzzer board (a small protoboard at the end of the buzzer cable)

Table 4. Buzzer board (a small protoboard at the end of the buzzer cable).

Reference	Function	Part, value and rating	Polarity or orientation
BZ1	Buzzer	Passive magnetic buzzer (a coil, not a piezo), 11.8 × 8.35 mm, coil about 40 Ω	Pin 1 positive, pin 2 negative
Q1	Buzzer transistor	BC337 NPN, TO-92: 45 V, 800 mA DC, 625 mW, gain 100 or more at 100 mA	The schematic numbers 1 collector, 2 base, 3 emitter; makers differ: find C, B and E with a tester before soldering
R7	Q1 base resistor	470 Ω	None
R8	Q1 base pull-down	10 kΩ	None
D2	Buzzer flyback diode	1N5819 Schottky, 40 V, 1 A	Pin 1 cathode on 5VDC, pin 2 anode on Q1′s collector

4.4 Presence sensor node

Table 5. Presence sensor node.

Reference	Function	Part, value and rating	Polarity or orientation
U4	Radar	HLK-LD2410C 24 GHz mmWave presence module; supply 5 to 12 V (5 V advised); logic 3.3 V	Pins 1 TX, 2 RX, 3 OUT, 4 GND, 5 VCC
C5	Bulk capacitor on 5VDC at the radar	10 µF, 10 V or more, polarized	Pin 1 positive, on 5VDC
C6	Decoupling capacitor on 5VDC at the radar	100 nF, 10 V or more	None
U13	Radar cable header at the sensor node	JST-XH, 5 pins	See chapter “Cables”

4.5 Cable ends

Table 6. Cable ends.

Reference	Cable	Kind
U22, U21	USB-C power cable, mains box end and bench node end	USB-C plugs
U23, U16	Bar-sense cable, mains box end and bench node end	JST-XH, 3 pins
U14, U12	Radar cable, bench node end and sensor node end	JST-XH, 5 pins
U17, U18	Screen cable, header end and screen end	JST-XH, 4 pins
U19	Keys cable, header end; the other end is soldered to the keys	JST-XH, 7 pins
U20	Buzzer cable, header end; the other end is soldered to the buzzer board	JST-XH, 3 pins

- Wire: all mains wiring is 14 AWG; all low-voltage wiring is 22 AWG.

5 The switching device (U1, Shelly 1PM Gen4)

U1, the Shelly 1PM Gen4, is a smart relay with its own power meter. It switches the power bar on and off, and it runs the lease script that turns the bar off by itself when the bench node stops renewing. It sits in the mains box, not inside the power bar.

5.1 Identity and ratings

Table 7. Identity and ratings.

Item	Value
Product	Shelly 1PM Gen4
Model code	`S4SW-001P16EU` (European model)
FCC ID	`2BDC6-1PMGEN4`
Supply	110–240 VAC, 50/60 Hz; or 24–30 VDC
Relay rating	16 A at 240 VAC; 10 A at 30 VDC
Operating temperature	−20 °C to 40 °C ambient
Firmware	2.1.0 (described in the Firmware Gospel)
Marks	FCC, CE, a μ, a house, a crossed-out bin, a lightning bolt in a triangle. No Canadian safety mark

- The supply here is 60 Hz, as everywhere on the North American grid; the Shelly is rated 50/60 Hz.

5.2 Inside the Shelly

- The relay: printed “HF7520 012-HSTP 16A 125VAC, 16A 250VAC, 8A 250VAC (COS ϕ = 0.4), TV-5 SF0816B1”. Every rating on it is in VAC.
- From the relay maker’s datasheet: HF7520/012-HSTP is a 12 VDC coil, one normally open contact (1 Form A), plastic sealed, AgSnO2 contacts, high-capacity type.
- Non-latching. One coil, two coil pins, and no latching option in the maker’s ordering scheme. So the contact opens whenever the coil loses drive: a Shelly reset with the coil undriven leaves the bar off.

5.3 Load types and the snubber

5.4 Wiring limits of the Shelly’s terminals

Table 8. Wiring limits of the Shelly’s terminals.

Maker document	Wire size	Other limits
“User and safety guide Shelly 1PM Gen4”	0.2 to 2.5 mm² (14 AWG at most)	One wire per terminal; insulation rated 105 °C or more
The maker’s web page	0.5 to 4 mm² (20 to 11 AWG)	Strip 6 to 7 mm; 0.4 Nm

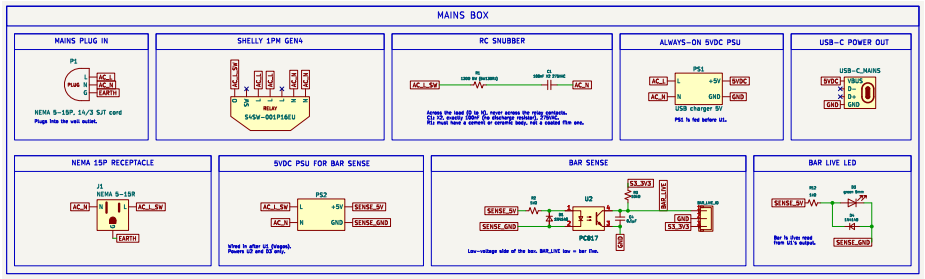
The two documents disagree on wire size. Wire of 0.5 to 2.5 mm² satisfies both. Use one wire per terminal and insulation rated 105 °C or more. The Shelly has no earth terminal.

5.5 Documents

- Shelly Knowledge Base, “Shelly 1PM Gen4”, https://kb.shelly.cloud/knowledge-base/shelly-1pm-gen4: “Inductive with RC Snubber (LED light drivers, transformers, fans, refrigerators, air-conditioners, washing machines, tumble dryers)”; “Max. switching current: 16A (240 V~)”; “Ambient working temperature: -20 °C to 40 °C”; wire 0.5 to 4 mm².
- Shelly, “User and safety guide Shelly 1PM Gen4”, https://kb.shelly.cloud/__attachments/a_e65c057158485c88982fe8b3ebb93ff571e6f1e66495eb531888c7e9f809dd93/Shelly%201PM%20Gen%204%20multilingual%20printed%20user%20and%20safety%20guide.pdf.
- Shelly RC Snubber accessory: https://us.shelly.com/products/rc-snubber.

6 The mains box

Figure 2. The mains box on the schematic.



Schematic box: MAINS BOX, in `hardware/KiCad/Power Protector.kicad_sch`. The mains box sits between the wall outlet and the power bar. It holds U1 (the Shelly), PS1 (the bench node supply), the snubber R1 and C1, and the bar-live sense. Its circuits follow in the order of the schematic’s boxes.

6.1 Layout

1. The mains box plugs into the wall receptacle by P1, its plug and cord.
2. U1, the Shelly, sits inside the box.
3. The power bar plugs into J1, the box’s receptacle, downstream of the Shelly.

6.2 MAINS PLUG IN (P1)

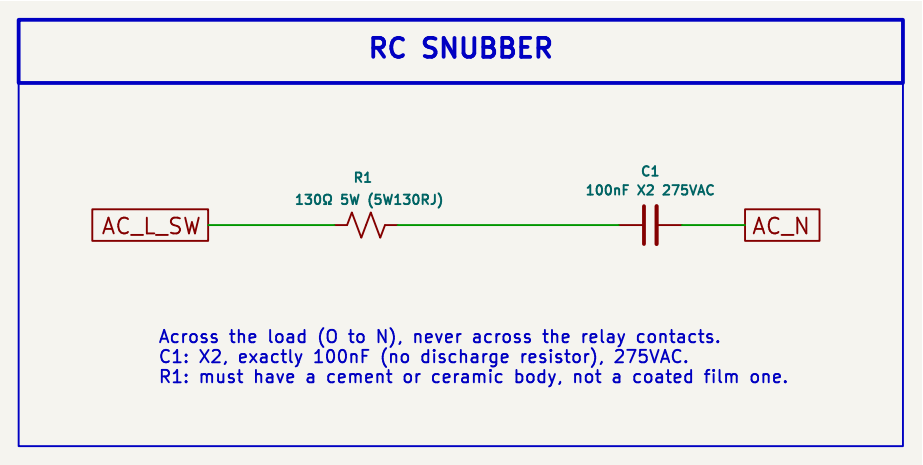
A NEMA 5-15P plug on 14/3 SJT cord. The cord is held by a clamp where it enters the box. In this build there is no separate strain relief, since the box is not moved; the clamp is part of the enclosure.

6.3 SHELLY 1PM GEN4 (U1)

U1’s L terminals take `AC_L`, its N terminals `AC_N` its output O carries the switched line `AC_L_SW`. SW is not connected. Two `AC_L` wires go to two of its three L terminals (pins 3 and 4); the third (pin 5) is not connected. Two `AC_N` wires go to its two N terminals (pins 6 and 7).

6.4 RC SNUBBER (R1, C1)

Figure 3. The RC snubber on the schematic.



R1, the snubber resistor (130 Ω, 5 W, ceramic or cement body, marked `5W130RJ`), in series with C1, the snubber capacitor (100 nF X2, marked 275 VAC), as discrete parts. The snubber runs from the Shelly's output (O, `AC_L_SW`) through R1, then C1, to neutral: across the load, never across the relay contacts. R1 must have a cement or ceramic body, not a coated film one. C1 is exactly 100 nF, with no discharge resistor.

6.5 ALWAYS-ON 5VDC PSU (PS1)

PS1, the bench node supply, is a 5 V USB charger. It is fed from the line before the Shelly, so the bench node outlives a cut. Any 5 VDC supply giving 1 A or more will do. The peak loads on `5VDC` add up to about 0.7 A: the devkit transmitting on WiFi, about 350 mA (340 mA peak in the ESP32-S3 datasheet); the radar, more than 200 mA; the buzzer through Q1, about 120 mA; the screen and the three key LEDs, some tens of mA.

6.6 USB-C POWER OUT (P3)

P3, the USB-C socket in the mains box wall. PS1's 5 V and ground go straight to its VBUS and GND; its D+ and D− are not connected. The USB-C cable to the bench node starts here.

6.7 NEMA 15P RECEPTACLE (J1)

J1, a NEMA 5-15R receptacle, fed by the switched line `AC_L_SW`, neutral `AC_N` and `EARTH`. The power bar plugs in here.

6.8 5VDC PSU FOR BAR SENSE (PS2)

Substitute for PS2: any unregulated transformer-type AC adapter of about 4.5 VDC that reads no more than about 7 V with no load, its positive output on `SENSE_5V`. Up to 7 V the LED current of U2 stays about 9× under its limit.

6.9 BAR SENSE

Figure 4. The bar sense on the schematic.

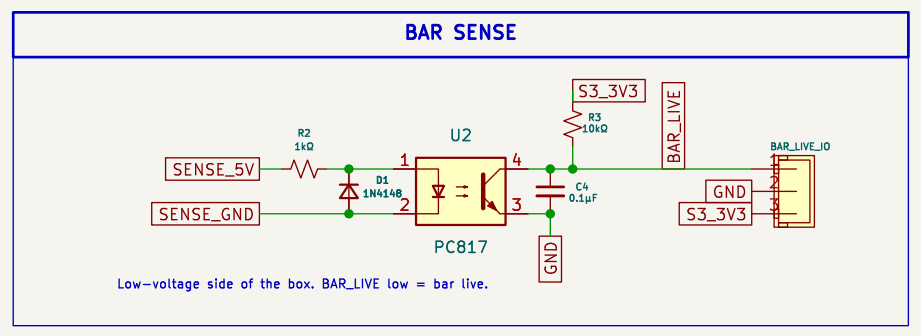
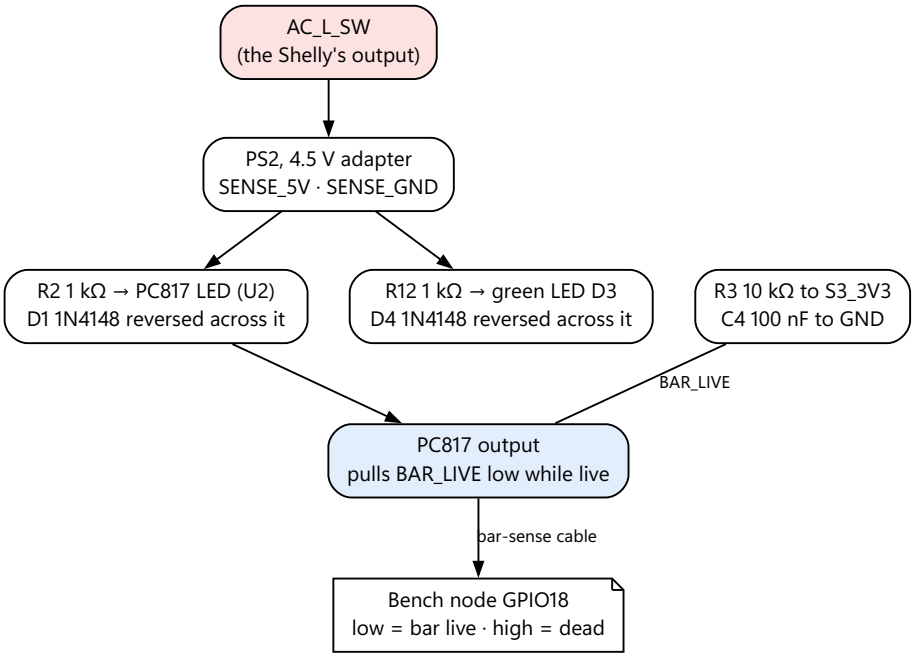


Figure 5. The bar-live sense, from the switched line to the bench node's pin.



1. PS2 drives the LED of U2, the PC817 optocoupler, through R2 (1 kΩ), with D1 (1N4148) reversed across that LED.
2. U2's output pulls `BAR_LIVE` low while the bar is live.
3. R3, a 10 kΩ pull-up to `S3_3V3`, and C4, a 100 nF capacitor to ground, sit on `BAR_LIVE` at the mains box end, on the box's low-voltage side. So `BAR_LIVE` is high when the bar is dead.
4. A 3-pin cable, about 10 ft (3 m) long, joins the mains box and the bench node, carrying `BAR_LIVE`, `GND` and `S3_3V3`.

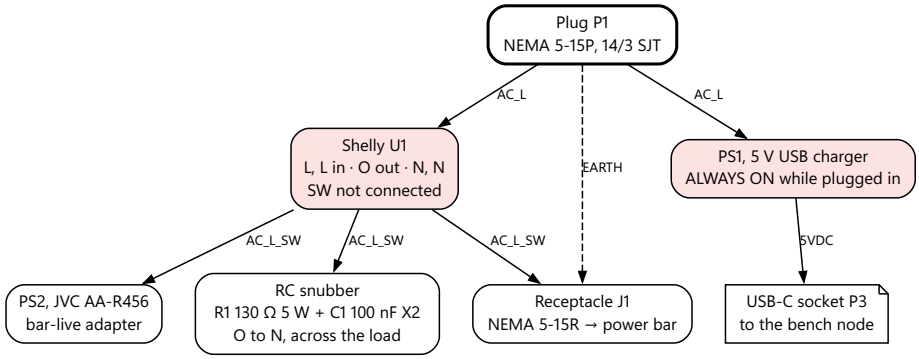
- PS2's side (`SENSE_5V`, `SENSE_GND`) touches nothing else: U2 isolates it from the bench node's side.
- `BAR_LIVE` low means the bar is live.

6.10 BAR LIVE LED (D3)

D3, a green 5 mm LED across PS2′s output, in the low-voltage part of the box, with R12 (1 kΩ) in series and D4 (1N4148) reversed across it. Lit, D3 shows that the Shelly's output is live.

6.11 Mains wiring

Figure 6. The mains wiring, from plug to receptacle.



All mains wiring is 14 AWG. As built, joints are made with Wago connectors (believed to be model 221); any connector that joins the conductors as drawn will do.

Table 9. Mains wiring.

Conductor	Runs to
EARTH	P1 and J1. Nothing else
AC_L (line, unswitched)	U1 (two L terminals) and PS1
AC_N (neutral)	U1 (two N terminals), PS1, PS2, J1 and the snubber (C1)
AC_L_SW (switched line, the Shelly's output O)	PS2, J1 and the snubber (R1)

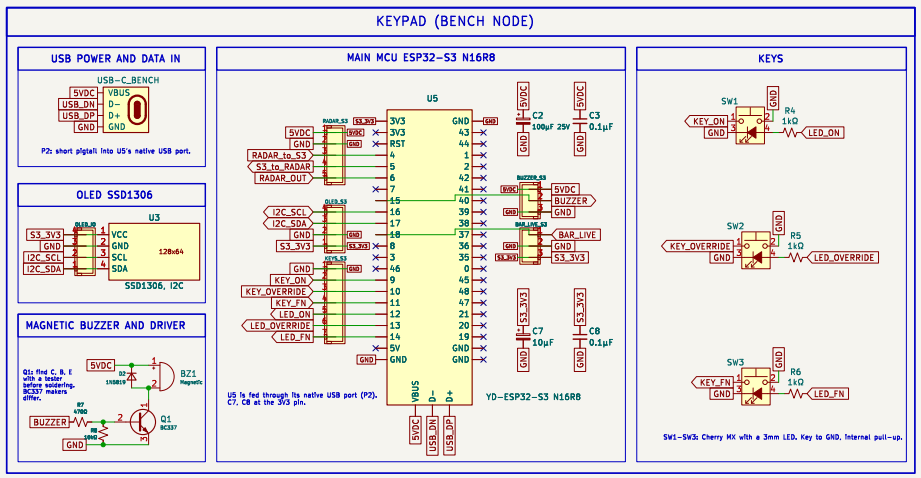
6.12 Separation

6.13 Heat

- The Shelly is rated to 40 °C ambient: the air inside the mains box must stay under 40 °C. The design takes room air as about 21 °C usually and 27–28 °C at the warmest, which leaves 12 to 19 °C for the temperature rise inside the box.

7 The bench node

Figure 7. The bench node on the schematic.



Schematic box: KEYPAD (BENCH NODE). The bench node makes the decisions. It sits at the bench and holds the three keys, the screen, and the buzzer on its cable. It is joined to the mains box by two cables (USB-C power and the 3-pin bar-sense cable) and to the presence sensor node by one.

7.1 USB POWER AND DATA IN (P2)

- A USB-C to USB-C cable runs from P3, the mains box USB-C socket, to P2, the bench node USB-C socket. It carries 5 V and ground only.
- P2 feeds U5′s native USB-C port by a short pigtail: 5 V, ground, and that port's data lines (USB_DP, USB_DN).
- U5′s 5V header pin and its second 3V3 header pin are not connected.
- No CC resistors are needed on either USB-C socket, and the boards fitted carry none. PS1′s 5 V is wired straight to VBUS and nothing negotiates power, so VBUS is live without them.
- PS1 must give 5 V DC at 1 A or more. The peak loads on 5VDC add up to about 0.7 A.

7.2 MAIN MCU ESP32-S3 N16R8 (U5)

- U5 is an ESP32-S3 devkit, N16R8 (16 MB flash, 8 MB octal PSRAM), with USB-C: the YD-ESP32-S3 board ("ESP32-S3-DevKitC-1 N16R8, USB-C (YD-ESP32-23 2022-V1.3)").
- Its IN-OUT jumper is open. With the jumper open, the header's 5V pin sits behind a diode on the devkit (marked D3 on the devkit maker's schematic; not the bar-live LED D3). So U5 is fed through its native USB port, from P2.
- C7 (10 µF) and C8 (100 nF) sit at U5′s 3V3 pin.
- As built, U5 sits on headers on a perfboard. Any construction will do, including wires soldered straight to the devkit.
- Devkit maker's schematic: YD-ESP32-S3 V1.4, https://raw.githubusercontent.com/rtek1000/YD-ESP32-23/main/YD-ESP32-S3-SCH-V1.4.pdf.

7.3 Pins of U5

The same names and numbers are in the pin file, `firmware/bench-node/firmware/src/pins.h`.

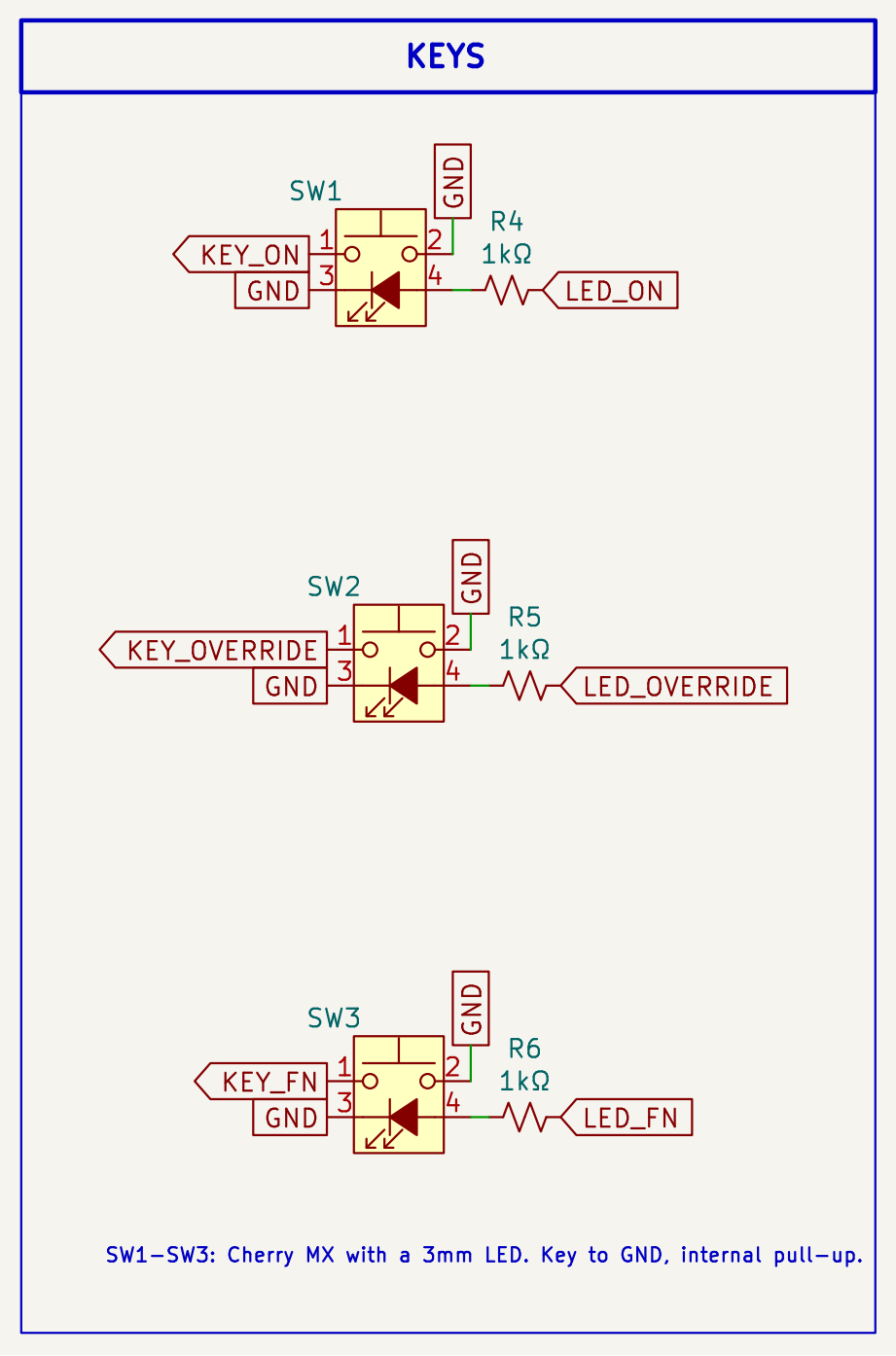
Table 10. Pins of U5.

Signal	GPIO	Goes to
`RADAR_to_S3`	GPIO4	U4, the radar, its TX
`S3_to_RADAR`	GPIO5	U4, the radar, its RX
`RADAR_OUT`	GPIO6	U4, the radar, its OUT pin
`KEY_ON`	GPIO9	SW1, the ON key
`KEY_OVERRIDE`	GPIO10	SW2, the OVERRIDE key
`KEY_FN`	GPIO11	SW3, the FN key
`LED_ON`	GPIO12	The LED in SW1, through R4
`LED_OVERRIDE`	GPIO13	The LED in SW2, through R5
`LED_FN`	GPIO14	The LED in SW3, through R6
`BUZZER`	GPIO15	Q1′s base through R7, on the buzzer board
`I2C_SCL`	GPIO16	U3, the screen
`I2C_SDA`	GPIO17	U3, the screen
`BAR_LIVE`	GPIO18	The bar-live sense in the mains box; low = bar live

None of these is a strapping, flash, PSRAM or USB pin of the ESP32-S3.

7.4 KEYS (SW1 to SW3)

Figure 8. The keys on the schematic.



- Three momentary switches, each with a yellow 3 mm LED: SW1 ON, SW2 OVERRIDE and SW3 FN. As built they are Cherry MX type; any momentary switch will do.
- Each key closes its `KEY_*` line to ground. No pull-up resistor is drawn; the firmware gives each key line an internal pull-up of the ESP32-S3 (described in the Firmware Gospel).

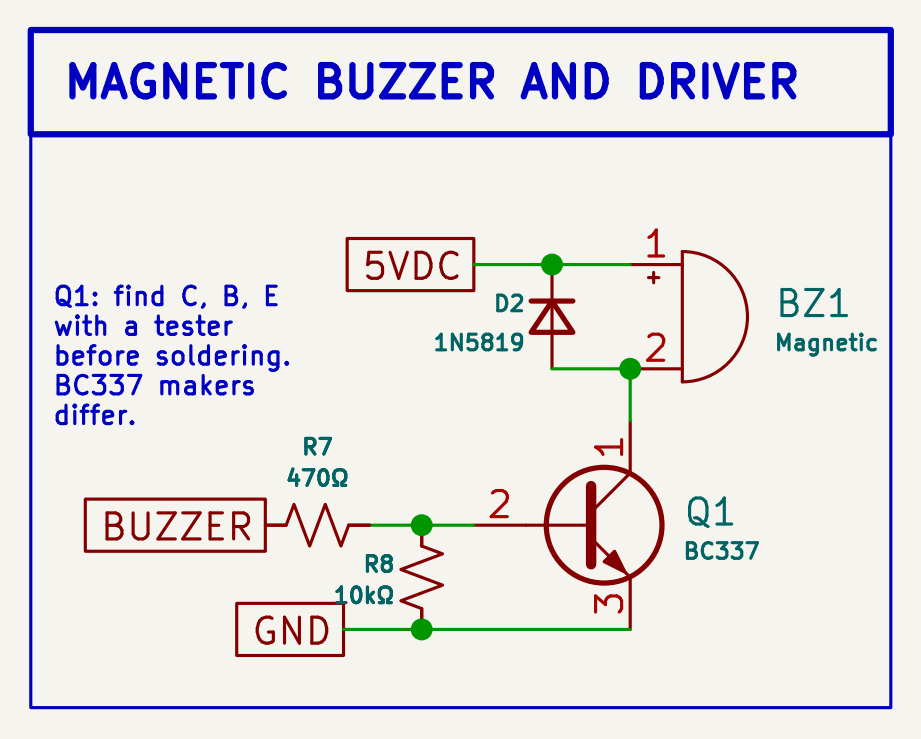
- Each key's LED runs from its `LED_*` line through 1 kΩ (R4, R5, R6) to the LED anode (key pin 4); the LED cathode (key pin 3) is on ground.
- All six key and LED signals, with ground, share one 7-pin connector, U10.

7.5 OLED SSD1306 (U3)

- U3 is a 0.96″ OLED, 128 × 64, SSD1306, I²C, on its own 4-pin cable. It runs on `S3_3V3`. On the schematic its symbol is `PP:OLED_0.96_I2C` and its value "0.96in SSD1306, I2C".
- Any 0.96″ 128 × 64 SSD1306 I²C module will do; such modules are common, and their colour does not matter.
- Screen boards differ in pin order: the cable must match the board in hand. Wire by the printed pin names.

7.6 MAGNETIC BUZZER AND DRIVER (BZ1, Q1)

Figure 9. The buzzer and its driver on the schematic.



- BZ1 is a passive magnetic buzzer (a coil, not a piezo), 11.8 × 8.35 mm. Its coil reads about 40 Ω.
- BZ1 is driven by Q1, a BC337 NPN transistor (TO-92), on the buzzer's own small protoboard at the end of the 3-wire buzzer cable:
  - `BUZZER` (GPIO15) feeds Q1's base through R7 (470 Ω); R8 (10 kΩ) pulls the base to ground.
  - BZ1 runs from `5VDC` (its + lead) to Q1's collector (its − lead). Q1's emitter is on ground.
  - D2, a 1N5819 Schottky diode, sits across BZ1, cathode on `5VDC`, to absorb the coil's turn-off spike.

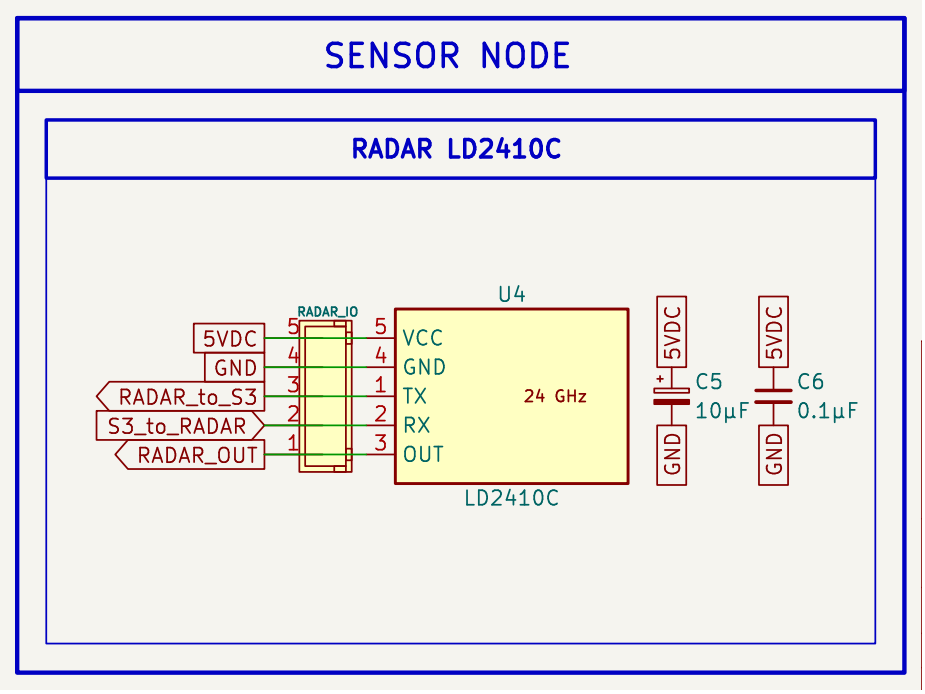
- BUZZER high turns Q1 on and BZ1 sounds.

> ⚠ CAUTION. BC337 makers number the leads differently. Find Q1′s collector, base and emitter with a tester before soldering. The base is the middle lead in every maker's numbering. The schematic numbers them 1 collector, 2 base, 3 emitter.

- Design checks (from the datasheets, not measured):
  - collector current about (5 V − 0.3 V) / 40 Ω ≈ 120 mA; base current about 5.5 mA, a forced gain of about 22, against a minimum gain of 100 for the lowest gain group: Q1 saturates with about 4.5× margin;
  - Q1 dissipates about 30 mW, about 84 mW at the datasheet's 0.7 V worst-case saturation voltage, against 625 mW rated;
  - GPIO15 sources about 5.5 mA, under its default 20 mA drive;
  - while U5 resets, GPIO15 floats and R8 holds Q1 off, so BZ1 stays silent through boot;
  - D2 clamps the spike to about 5 V plus its forward drop, well under Q1′s 45 V rating.
- Any small passive magnetic buzzer will do; such buzzers are common, and their exact size and rating do not matter. A 16 Ω coil in place of the 40 Ω one might not saturate Q1 in the worst case and would sound quieter; that is not a safety issue.
- Why a driver: wired straight from GPIO15, a 40 Ω coil draws about 80 mA, above the pin's rating, and its turn-off spike has no diode. Wired that way, the bench node restarts whenever a sound changes pitch while the pin is driven.

8 The presence sensor node

Figure 10. The presence sensor node on the schematic.



Schematic boxes: SENSOR NODE and RADAR LD2410C. The presence sensor node is a small separate unit holding U4, the radar that tells the bench node someone is there.

- Sensor: U4, an HLK-LD2410C 24 GHz mmWave presence module, in its own small node, aimed to see only the bench and its chair, and nothing beyond.
- Cable: a 5-pin cable to the bench node, about 1.5 m, made from USB-type cable with a foil shield. The shield is tied to ground at the bench node end only.
- C5 (10 µF) and C6 (100 nF) sit on `5VDC` at the sensor.
- From the maker's manual: pins 1 TX, 2 RX, 3 OUT, 4 GND, 5 VCC; supply 5 to 12 V (5 V advised); logic at 3.3 V; serial port at 256 000 baud by default.
- U4 runs at 5 V, exactly the advised value. Its 3.3 V signals go straight to U5′s 3.3 V pins, with no level shifting.
- How the radar is tuned and filtered is firmware (described in the Firmware Gospel).

9 The schematic

- File: `hardware/KiCad/Power Protector.kicad_sch`, revision A, KiCad 9. The sheet is A3. Its symbols are in the project library `hardware/KiCad/PP.kicad_sym`, so it opens on another computer.

- Its PDF, `hardware/KiCad/Power Protector - Schematic.pdf`, is the A3 sheet fitted onto one US Letter page in landscape (scaled 0.665). Its smallest text prints at about 2 pt, so read small pin labels on a screen, zoomed.
- Its bill of materials, `hardware/Deadman - BoM.csv`, lists the parts of this file and the schematic unit by unit (mains box, bench node, sensor node) and for the cables, with quantities and references. It belongs to revision A, with the schematic.
- Title block: revision "A", date "2026-10-25".
- SVG export: `docs/figures/power-protector-schematic.svg`, the whole sheet. One SVG per box of the schematic sits beside it, named `docs/figures/schematic-<box>.svg` (for example `schematic-shelly.svg`, `schematic-bar-sense.svg`, `schematic-mcu.svg`).
- KiCad's electrical rules check: 0 errors, 2 warnings, both "unconnected wire endpoint" on the two short stubs marked "Cable S/F" (the cable shields).
- Connectors: every header and cable end U6 to U20 and U23 is JST-XH. P1, J1 and the USB-C sockets and cable ends (P2, P3, U21, U22) are not JST. The JST connectors are a convenience; any connector that keeps the same connections will do.
- JST pin numbers are a schematic convention. Each cable end is numbered to match its mating header. Which physical pin is pin 1 is left to the builder: choose it on each header, and number its cable end the same way, so that pin k meets pin k.
- The schematic is a visual aid; chapter "Every connection" states every connection in words.

10 Every connection

This is the Deadman Power Protector's wiring, net by net, as drawn in `hardware/KiCad/Power Protector.kicad_sch`.

10.1 Part pins used in the net tables

- U1, the Shelly, pins 1 to 7, left to right: O, SW, L, L, L, N, N.
- U2, the PC817: 1 anode, 2 cathode, 3 emitter, 4 collector.
- Each key (SW1 to SW3): pins 1 and 2 the switch contact; 4 the LED anode; 3 the LED cathode.
- Polarized capacitors (C2, C5, C7): pin 1 positive.
- Each wall adapter (PS1, PS2): 1 line in, 2 neutral in, 3 positive out, 4 negative out.
- Q1, the BC337: 1 collector, 2 base, 3 emitter, as drawn (find the leads with a tester; makers differ).
- BZ1, the buzzer: 1 positive, 2 negative. D2: 1 cathode, 2 anode.
- U5, the devkit, has, besides its header pins, three pins for its native USB-C port: VBUS, D−, D+.

10.2 Mains nets (14 AWG)

Table 11. Mains nets (14 AWG).

Net	Every pin on it
AC_L (line, unswitched)	P1 L; U1 pins 3 and 4 (L, L); PS1 pin 1
AC_N (neutral)	P1 N; J1 N; U1 pins 6 and 7 (N, N); PS1 pin 2; PS2 pin 2; C1 pin 2
AC_L_SW (switched line)	U1 pin 1 (O); J1 L; PS2 pin 1; R1 pin 1
EARTH	P1 ground; J1 ground. Nothing else
Snubber middle (no name)	R1 pin 2; C1 pin 1
Not connected	U1 pin 2 (SW); U1 pin 5 (the third L)

10.3 Bar-sense nets in the mains box (22 AWG)

Table 12. Bar-sense nets in the mains box (22 AWG).

Net	Every pin on it
SENSE_5V (PS2′s positive output)	PS2 pin 3; R2 pin 1; R12 pin 1
SENSE_GND (PS2′s negative output)	PS2 pin 4; U2 pin 2 (cathode); D1 anode; D3 cathode; D4 anode
U2 LED anode (no name)	R2 pin 2; U2 pin 1 (anode); D1 cathode
D3 anode (no name)	R12 pin 2; D3 anode; D4 cathode

10.4 Low-voltage nets (22 AWG)

Each net lists every pin on it. Cable ends carry the same signal as the header pin they mate with.

Table 13. Low-voltage nets (22 AWG).

Net	Every pin on it
5VDC	PS1 pin 3; P3 VBUS; USB cable ends U22 and U21 VBUS; P2 VBUS; U5 native-port VBUS; C2 pin 1; C3 pin 1; U6 pin 1; U14 pin 1; U12 pin 5; U13 pin 5; U4 pin 5 (VCC); C5 pin 1; C6 pin 1; U7 pin 1; U20 pin 1; BZ1 pin 1 (+); D2 pin 1 (cathode)
GND	PS1 pin 4; P3 GND; both USB cable ends' GND; P2 GND; U5 GND (two header pins); C2, C3, C4, C5, C6, C7, C8 pin 2; SW1, SW2, SW3 pins 2 and 3; U3 GND; U4 pin 4 (GND); U2 pin 3 (emitter); Q1 pin 3 (emitter); R8 pin 2; U6 pin 2, U7 pin 3, U8 pin 2, U9 pin 2, U10 pin 1, U11 pin 2, U13 pin 4, U15 pin 3, and each of their cable ends' matching pin
S3_3V3	U5 3V3 (one header pin); C7 pin 1; C8 pin 1; U9 pin 1; U17 pin 1; U18 pin 4; U15 pin 4; U3 VCC; U8 pin 3; U16 pin 3; U23 pin 3; U11 pin 3; R3 pin 1
BAR_LIVE	U2 pin 4 (collector); R3 pin 2; C4 pin 1; U11 pin 1; U23 pin 1; U16 pin 1; U8 pin 1; U5 GPIO18
USB_DP	P2 D+; U5 native-port D+
USB_DN	P2 D−; U5 native-port D−
I2C_SDA	U5 GPIO17; U9 pin 3; U17 pin 3; U18 pin 1; U15 pin 1; U3 SDA
I2C_SCL	U5 GPIO16; U9 pin 4; U17 pin 4; U18 pin 2; U15 pin 2; U3 SCL
BUZZER	U5 GPIO15; U7 pin 2; U20 pin 2; R7 pin 1
Q1 base (no name)	R7 pin 2; R8 pin 1; Q1 pin 2
Q1 collector (no name)	Q1 pin 1; BZ1 pin 2 (−); D2 pin 2 (anode)
KEY_ON	U5 GPIO9; U10 pin 2; U19 pin 2; SW1 pin 1
KEY_OVERRIDE	U5 GPIO10; U10 pin 3; U19 pin 3; SW2 pin 1
KEY_FN	U5 GPIO11; U10 pin 4; U19 pin 4; SW3 pin 1
LED_ON	U5 GPIO12; U10 pin 5; U19 pin 5; R4 pin 2. R4 pin 1 to SW1 pin 4 (LED anode)
LED_OVERRIDE	U5 GPIO13; U10 pin 6; U19 pin 6; R5 pin 2. R5 pin 1 to SW2 pin 4 (LED anode)
LED_FN	U5 GPIO14; U10 pin 7; U19 pin 7; R6 pin 2. R6 pin 1 to SW3 pin 4 (LED anode)
RADAR_to_S3	U5 GPIO4; U6 pin 3; U14 pin 3; U12 pin 3; U13 pin 3; U4 pin 1 (TX)
S3_to_RADAR	U5 GPIO5; U6 pin 4; U14 pin 4; U12 pin 2; U13 pin 2; U4 pin 2 (RX)
RADAR_OUT	U5 GPIO6; U6 pin 5; U14 pin 5; U12 pin 1; U13 pin 1; U4 pin 3 (OUT)
Not connected	P3 D+ and D−; both USB cable ends' D+ and D−; U5 5V header pin; U5 second 3V3 header pin; every U5 pin not named in this table

⚠ CAUTION. The Deadman Power Protector's radar cable is not pin for pin by position: pin 1 at the bench node (U6) is 5VDC, but pin 1 at the sensor node (U13) is RADAR_OUT. Follow the net names, not the pin numbers.

11 Cables

Six cables join the Deadman Power Protector's units and the parts on the bench node. Each cable end carries, on its pin k, the same signal as pin k of the header it plugs into. The two ends of one cable are not always numbered alike: on the radar cable, pin 1 is 5VDC at the bench node but RADAR_OUT at the sensor node. Follow the signal names.

Table 14. Cables.

Cable	From	To	Ends	Length	Kind
USB-C power	P3 (mains box)	P2 (bench node)	U22, U21	About 10 ft (3 m), the same distance as the bar-sense cable; exact length is the builder's choice	USB-C to USB-C cable; 5 V and ground only, D+ and D− not carried
Bar sense	U11 (mains box)	U8 (bench node)	U23, U16	About 10 ft (3 m)	Cat7 S/FTP; shield tied to ground at the bench node end only. Which Cat7 conductors carry which signal is the builder's choice
Radar	U6 (bench node)	U13 (sensor node)	U14, U12	About 1.5 m	USB-type cable with foil shield; shield tied to ground at the bench node end only
Screen	U9	U15 (at the screen)	U17, U18	Under 12 cm	Individual 22 AWG wires
Keys	U10	The keys	U19	Under 12 cm	Individual 22 AWG wires, soldered straight to the keys and their LEDs
Buzzer	U7	The buzzer board	U20	Under 12 cm	Individual 22 AWG wires, soldered straight to the buzzer board

Inside each box, every wire is under 12 cm.

11.1 USB-C power cable (schematic box BENCH NODE TO MAINS BOX)

Figure 11. The USB-C power cable.

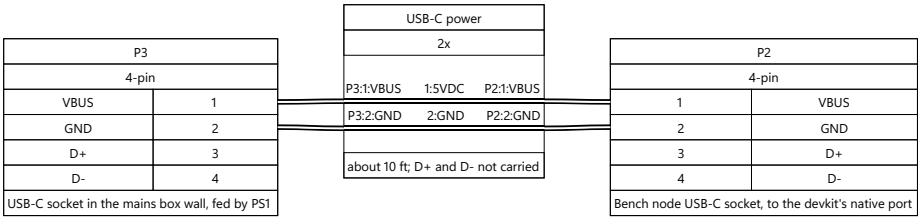


Table 15. USB-C power cable (schematic box BENCH NODE TO MAINS BOX).

P3 / U22 (mains box)	Signal	U21 / P2 (bench node)
VBUS	5VDC	VBUS
GND	GND	GND
D+, D−	Not carried	D+, D−

11.2 Bar-sense cable, 3-pin (schematic box BENCH NODE TO MAINS BOX)

Figure 12. The bar-sense cable.

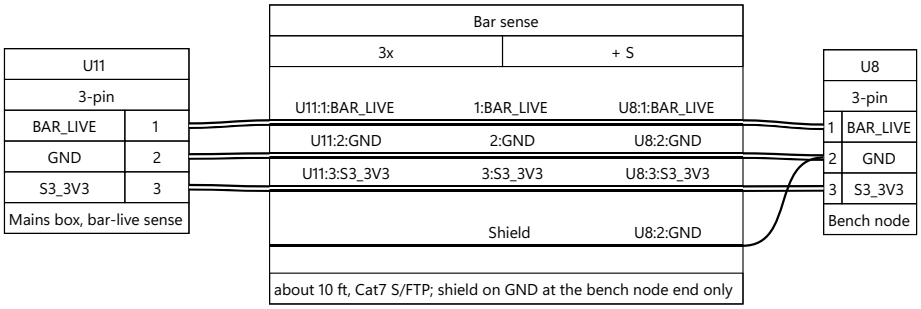


Table 16. Bar-sense cable, 3-pin (schematic box BENCH NODE TO MAINS BOX).

U11 / U23 pin (mains box)	Signal	U16 / U8 pin (bench node)
1	BAR_LIVE	1
2	GND	2
3	S3_3V3	3
Shield	Tied to GND at the bench node end only	—

11.3 Radar cable, 5-pin (schematic box BENCH NODE TO LD2410C RADAR)

Figure 13. The radar cable.

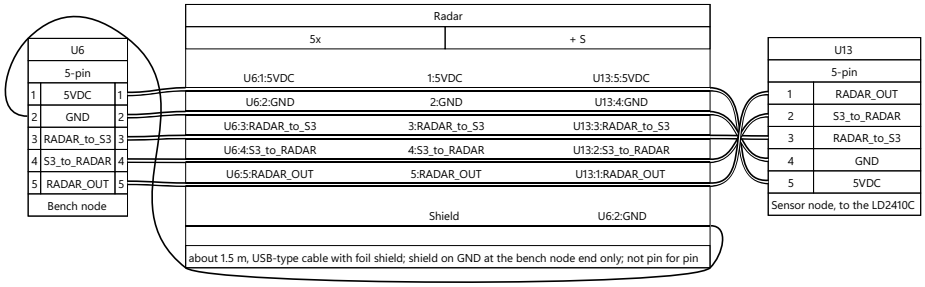


Table 17. Radar cable, 5-pin (schematic box BENCH NODE TO LD2410C RADAR).

U6 / U14 pin (bench node)	Signal	U12 / U13 pin (sensor node)	U4 (radar) pin
1	5VDC	5	5 (VCC)
2	GND	4	4 (GND)
3	RADAR_to_S3	3	1 (TX)
4	S3_to_RADAR	2	2 (RX)
5	RADAR_OUT	1	3 (OUT)
Shield	Tied to GND at the bench node end only	—	—

11.4 Screen cable, 4-pin (schematic box BENCH NODE - S3 to OLED)

Figure 14. The screen cable.

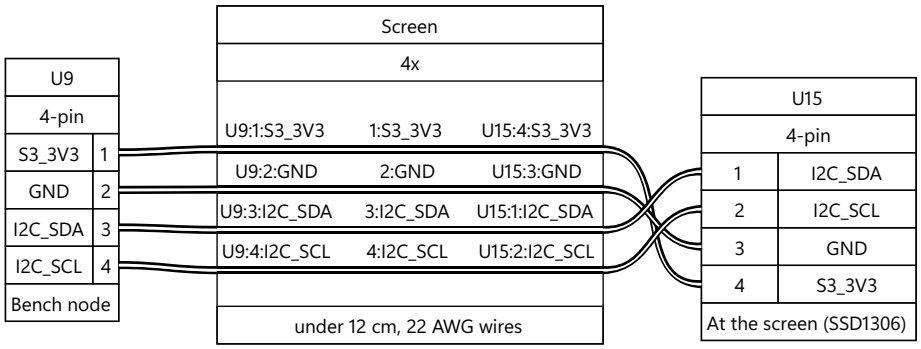


Table 18. Screen cable, 4-pin (schematic box BENCH NODE - S3 to OLED).

U9 / U17 pin (bench node)	Signal	U18 / U15 pin (at the screen)
1	S3_3V3	4
2	GND	3
3	I2C_SDA	1
4	I2C_SCL	2

11.5 Keys cable, 7-pin (schematic box BENCH NODE - S3 to KEYS and LEDS)

Figure 15. The keys cable.

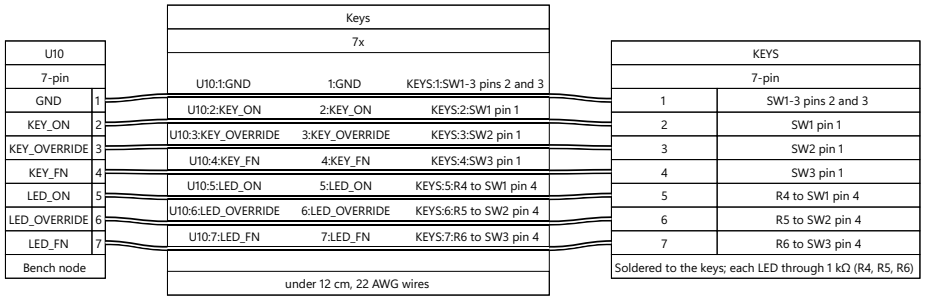


Table 19. Keys cable, 7-pin (schematic box BENCH NODE - S3 to KEYS and LEDS).

U10 / U19 pin	Signal	At the keys
1	GND	SW1, SW2, SW3 pins 2 and 3
2	KEY_ON	SW1 pin 1
3	KEY_OVERRIDE	SW2 pin 1
4	KEY_FN	SW3 pin 1
5	LED_ON	R4 to SW1 pin 4
6	LED_OVERRIDE	R5 to SW2 pin 4
7	LED_FN	R6 to SW3 pin 4

11.6 Buzzer cable, 3-pin (schematic box BENCH NODE - S3 to BUZZER)

Figure 16. The buzzer cable.

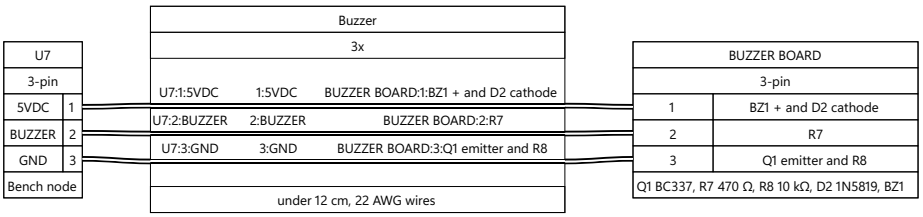


Table 20. Buzzer cable, 3-pin (schematic box BENCH NODE - S3 to BUZZER).

U7 / U20 pin	Signal	At the buzzer board
1	5VDC	BZ1 +, D2 cathode
2	BUZZER	R7
3	GND	Q1 emitter, R8

12 Enclosures

The enclosures of the Deadman Power Protector are not part of the build: their form is the builder's choice. The mains box enclosure must meet these requirements:

- Mains and low-voltage wiring share it, kept at least 3 cm apart, and no wire inside can move.
- PS1 and PS2, the two power supplies, are fixed in place.
- The cord P1 is held by a clamp where it enters the box.
- P3, the USB-C socket, sits in the box wall.
- The air inside must stay under 40 °C, the Shelly's ambient rating.
- D3, the bar-live LED, sits in the low-voltage part of the box.

13 Substitutes and safe changes

13.1 What a builder may substitute

Table 21. What a builder may substitute.

Part	May be replaced by
The Wago connectors in the mains box	Any connector that joins the conductors as drawn
PS1, the bench node supply	Any 5 VDC supply giving 1 A or more
PS2, the bar-sense adapter	Any unregulated transformer-type AC adapter of about 4.5 VDC, no more than about 7 V with no load, positive on `SENSE_5V`
BZ1, the buzzer	Any small passive magnetic buzzer
U3, the screen	Any 0.96″ 128 × 64 SSD1306 I²C module; match the cable to its pin order
The enclosures	Anything that meets the requirements in chapter “Enclosures”
The perfboard and headers under U5	Wires soldered straight to the devkit
The Cherry MX keys	Any momentary switches
The JST connectors	Any connector that keeps the same connections

13.2 What may change, and what each change touches

Table 22. What may change, and what each change touches.

Change	Allowed when	It touches
A signal's GPIO	The new pin is not a strapping pin (GPIO0, 3, 45, 46), not a flash or PSRAM pin (GPIO26 to 37), not the native USB pair (GPIO19, 20), and not already used	The pin file `firmware/bench-node/firmware/src/pins.h` and a firmware rebuild; the schematic; a re-test of that signal's path
R2, the optocoupler LED resistor	U2′s LED current stays well under 50 mA, and at about 1 mA or more, enough for 0.33 mA out at the lowest transfer ratio	Nothing else
R3 and C4, the pull-up and filter on `BAR_LIVE`	The pull-up still goes to 3.3 V	The bar-sense timing the firmware relies on: re-test the bar sense
R7 and R8, Q1′s base resistors	Q1 still saturates: base current about 1/20 of the buzzer current or more	Nothing else
R4 to R6 and R12, the LED resistors	Brightness only	Nothing else
Q1, the buzzer transistor	Any small NPN rated 45 V and 0.5 A or more, with a gain of 100 or more at 100 mA. Check its lead order	Nothing else
D2, the buzzer flyback diode	Any Schottky or 1N400x-type diode rated 1 A and 40 V or more, cathode on `5VDC`	Nothing else
U2, the optocoupler	Any 4-pin phototransistor optocoupler with 5 kV isolation and a transfer ratio of 50 % or more	Nothing else
Connectors, enclosures, how the boards are built	Anything that keeps the same connections	Nothing else

A change on the cut path or the warning path (`BAR_LIVE`, the buzzer, the keys, the screen, the radar, the lease) is re-tested on that path before the Deadman Power Protector goes back into use.

The list of what must never change in the mains box is in chapter "Safety".

14 Building and checks

- Order of assembly and checks before energizing: left to the builder.
- Voltages at each node: the rails are named for their voltage: `5VDC` (5 V), `S3_3V3` (3.3 V), `SENSE_5V` (PS2′s output, about 4.5 V nominal, up to about 7 V with no load).

- Which Cat7 conductors carry which signal, and the USB-C cable's exact length: the builder's choice.

14.1 Expected values

Table 23. Expected values.

Check	Expected	How to check
`BAR_LIVE` after the Shelly turns off, no load on the bar	Goes dead (high) about 7 s after the Shelly turns off; so a cut that lasts only 5 s before the bar is switched back on never shows as dead	The bench node's console log (described in the Firmware Gospel)
`BAR_LIVE` after the Shelly turns on	Goes live (low) within about 13 ms	The bench node's console log
D3, the bar-live LED	Lit while the Shelly's output is live	Look at it
Shelly internal temperature, one soldering iron (about 0.4 A at 120 V) for an hour or more, room about 23 °C	About 45 °C. That reading is inside the Shelly's case, above the air around it; the box air is taken as under 40 °C	`bar.t` in the bench node's `/api/state`, or `shelly_temp_c` in its MQTT state message (both described in the Firmware Gospel)
BZ1 through U5′s boot	Silent: R8 holds Q1 off while GPIO15 floats	A design check from the datasheets, not measured
KiCad electrical rules check of the schematic	0 errors, 2 warnings ("unconnected wire endpoint" on the two "Cable S/F" shield stubs)	KiCad 9.0.7

15 Known problems, maintenance and end of life

15.1 Known problems

Table 24. Known problems.

What you see	What it means	Fix
D3, the green bar-live LED, is lit	PS2 is powered: the Shelly's output is live	—
The bar is on but `BAR_LIVE` reads dead at the bench node	A bad crimp on the `BAR_LIVE` wire of the 3-pin bar-sense cable does exactly this	Redo the crimp; then check R3 and U2
`BAR_LIVE` reads live long after a cut	Normal for about 7 s (no bleeder across PS2). Much longer: the Shelly's relay may still be closed	Check the Shelly; the alarm is described in the Firmware Gospel
`BAR_LIVE` always reads live (low), even with the bar off	GPIO18 is held to ground. A solder bridge from GPIO18 to ground on the bench node does exactly this	Remove the bridge
The bench node restarts when the buzzer changes pitch	BZ1 is not driven through Q1	On the buzzer board, Q1, R7, R8 and D2 must be fitted
BZ1 is silent	Q1′s leads may be swapped	Find Q1′s collector, base and emitter with a tester

15.2 Maintenance and end of life

16 Datasheet limits

Only the limits of parts used in the Deadman Power Protector, from the makers' documents. They describe the catalogue part, not a measurement of the one fitted. Where the fitted part's maker is unknown, a representative sheet is named, and the lowest figure among makers is the one that holds. "As used" figures follow from the schematic. Every part is used with margin.

16.1 The Shelly's relay, Hongfa HF7520/012-HSTP

Table 25. The Shelly's relay, Hongfa HF7520/012-HSTP.

Item	Value	Page
Coil, 12 VDC, 1 Form A	Pick-up 9.6 VDC max.; drop-out 1.2 VDC min.; 720 Ω ±10 %	174
Release time	5 ms max. at rated voltage	173
Dielectric strength, coil to contacts	2500 VAC for 1 min	173
Safety approvals, UL/CUL file E133481, high-capacity type	"16A 125VAC 85°C HP type"; "13A 125VAC 105°C HP type"; "TV-5 120VAC HP type"	173
Safety approvals, TÜV, high-capacity type	"16A 250VAC HP type"; "8A 250VAC cosø=0.4 HP type"	173

The relay's printed "16A 125VAC" matches the UL/CUL rating "16A 125VAC 85°C HP type".

16.2 The Shelly 1PM Gen4 (U1)

The Shelly's own limits are in chapter "The switching device (U1, Shelly 1PM Gen4)": supply 110–240 VAC 50/60 Hz, relay 16 A at 240 VAC, –20 °C to 40 °C ambient, wire 0.5 to 2.5 mm² with 105 °C insulation, inductive loads only with an RC snubber.

16.3 The radar, HLK-LD2410C (U4)

From Hi-Link, "HLK-LD2410C Human Presence Motion Module User Manual", V1.00, revised 2022-11-7 (https://naylampmechatronics.com/img/cms/001080/HLK-LD2410C_datasheet.pdf):

Table 26. The radar, HLK-LD2410C (U4).

Item	Value	Page
Pins	"1 UART_Tx, 2 UART_Rx, 3 OUT, 4 GND, 5 VCC"	7
Supply	"Power input 5~12V（advise 5V）"	7
Logic	"The module IO output level is 3.3V. The default baud rate of the serial port is 256000"	8
Supply capacity	"the power supply capacity of the input power supply is required to be greater than 200mA"	8, §5.1
Supply current	"Average operating current \	79 mA"; about 130 mA at most (Fig. 11)
Detection	"Detection distance \	0.75m ~ 6m, adjustable"; "Distance resolution \
Ambient temperature	"-40 ~ 85℃"	17

As used: 5 V, exactly the advised value. Its 3.3 V signals go straight to 3.3 V pins, with no level shifting. The firmware caps the range well inside 0.75 to 6 m (described in the Firmware Gospel).

16.4 The ESP32-S3 (U5)

Table 27. The ESP32-S3 (U5).

Item	Value	Source
Strapping pins	“GPIO0, GPIO3, GPIO45, GPIO46 : Strapping pins”	Espressif, “ESP32-S3 Series Datasheet” v2.2, §2.3.5
PSRAM pins (octal PSRAM modules)	“pins IO35, IO36, and IO37 are connected to the Octal SPI PSRAM and are not available for other uses”	Espressif, “ESP32-S3-WROOM-1/-1U Datasheet” v1.8, Table 3-1, note b
Flash and PSRAM pins of the N16R8	GPIO26 to GPIO37: “Do not use the pins connected to in-package flash/PSRAM for any other purposes.”	Espressif, “ESP32-S3 Series Datasheet” v2.2, Table 2-14
Default pin drive	“GPIO17 and GPIO18: 10 mA … All other pins: 20 mA.”	Espressif, “ESP32-S3 Series Datasheet” v2.2, p. 18, Table 2-1, footnote 5
Strongest pin drive	40 mA source, 28 mA sink	The same, Table 5-4
Supply	“Allowed input voltage \	-0.3 \
Logic levels at 3.3 V	Input high 0.75 × VDD or more, input low 0.25 × VDD or less; output high 0.8 × VDD or more, output low 0.1 × VDD or less	The same, Table 5-4
Internal pull-up	45 kΩ typical	The same, Table 5-4
WiFi transmit peak	“802.11b, 1 Mbps, @21 dBm \	340″ mA
Operating temperature, N16R8 module	−40 to 65 °C	Espressif, “ESP32-S3-WROOM-1/-1U Datasheet” v1.8, Table 1-1

As used: GPIO4, 5, 6 and 9 to 18. None is a strapping, flash or PSRAM pin. The heaviest pin load is GPIO15′s about 5.5 mA into Q1′s base, against the 20 mA default drive.

Documents: Espressif, “ESP32-S3 Series Datasheet” v2.2, https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf; Espressif, “ESP32-S3-WROOM-1 & ESP32-S3-WROOM-1U Datasheet” v1.8, https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf. Both addresses serve the maker’s latest edition.

16.5 The 1N5819 (D2)

From onsemi, “1N5817, 1N5818, 1N5819”, 1N5817/D Rev. 11, August 2021 (https://www.onsemi.com/pdf/datasheet/1n5817-d.pdf), p. 2: reverse voltage

VRRM “40 V”; “Average Rectified Forward Current ... IO | 1.0 | A”; surge IFSM “25 (for one cycle) | A”; forward voltage “0.34 / 0.6 / 0.9 | V” at 0.1 / 1.0 / 3.0 A.

As used (D2, the buzzer flyback diode): 5 V in reverse, 8× under 40 V; about 120 mA at each turn-off of the buzzer coil, about 8× under 1 A.

16.6 The BC337 (Q1)

Table 28. The BC337 (Q1).

Item	Value	Source
Lead order	“C(1) B(2) E(3)”	Diotec BC337 datasheet
Lead order	Pin 1 emitter, pin 2 base, pin 3 collector	Philips Semiconductors, “BC817; BC817W; BC337”, Rev. 05, 2005-01-21, Table 3 (SOT54, the TO-92 body)
Minimum gain, lowest group (-16)	100 at 100 mA	Diotec BC337 datasheet, p. 2
Dissipation	625 mW	Diotec BC337 datasheet
Collector-emitter voltage	45 V	Diotec BC337 datasheet
Lead order, emitter-base voltage, collector current, gain groups, saturation	“1 collector, 2 base, 3 emitter”; VEBO 5 V; IC 800 mA DC; gain groups 16, 25, 40 at 100–250, 160–400, 250–630 (VCE = 1 V, IC = 100 mA); VCE(sat) 0.7 V max at IC = 500 mA, IB = 50 mA	onsemi, “BC337”, BC337-FSC/D Rev. 3, April 2024, pp. 1–2; Diotec, “BC337...BC338”, version 2026-03-30 (they agree on every figure)

The Philips sheet numbers the outer leads the other way from onsemi and Diotec, so Q1′s collector, base and emitter are found with a tester before soldering.

Documents: onsemi, “BC337”, BC337-FSC/D Rev. 3, https://www.onsemi.com/download/data-sheet/pdf/bc337-fsc-d.pdf; Diotec, “BC337...BC338”, version 2026-03-30, https://diotec.com/tl_files/diotec/files/pdf/datasheets/bc337.pdf; Philips Semiconductors, “BC817; BC817W; BC337”, Rev. 05, 21 January 2005, document number 9397 750 14022, in a reseller’s copy: https://www.handsontec.com/pdf_files/BC337_5.pdf.

As used (Q1, the buzzer transistor): about 120 mA collector current; base current about 5.5 mA through R7 (470 Ω), a forced gain of about 22 against at least 100 for the lowest group: Q1 saturates with about 4.5× margin. 5 V on the collector, 9× under 45 V. Dissipation, taking 0.7 V as a worst case: about 84 mW, 7× under 625 mW. VCE(sat) at exactly 120 mA: not given by either sheet.

16.7 The optocoupler, PC817 (U2)

The fitted part’s maker is unknown. Sheets: Sharp, “PC817XxNSZ1B Series”, Sheet No. OP18002EN, 2018-01-15, p. 4 (https://global.sharp/products/device/lineup/data/pdf/datasheet/PC817XxNSZ1B_e.pdf); Everlight, “EL817

Series”, DPC-0000046 Rev. 20, 2025-01-20 (https://www.everlighteurope.com/custom/files/datasheets/DPC-0000046.pdf); Lite-On, “LTV-817/827/847 Series”, BNS-OD-C131/A4, January 2010, in a distributor’s copy (https://media.digikey.com/pdf/data%20sheets/lite-on%20pdfs/ltv-817_827_847(m,s,s_ta1).pdf).

Table 29. The optocoupler, PC817 (U2).

Limit	Sheets say	As used	Margin
LED forward current	Sharp “Forward current \	IF \	50 \
LED reverse voltage	6 V on all three	D1 (1N4148), reversed across the LED, limits it to about 1 V	About 6×
Transfer ratio	Sharp “Ic \	IF=5mA,VCE=5V \	MIN.2.5 \
Collector-emitter voltage	Sharp 80 V; Everlight and Lite-On 35 V	3.3 V at most	About 10× under 35 V
Isolation	5 kV rms on all three (Sharp: “AC for 1 minute”)	Between PS2′s 4.5 V side and the bench node’s 3.3 V	—

16.8 The 1N4148 (D1, D4)

From onsemi, “1N91x, 1N4x48 … Small Signal Diode”, 1N914/D Rev. 6, September 2024 (https://www.onsemi.com/download/data-sheet/pdf/1n914-d.pdf), p. 2: “Maximum Repetitive Reverse Voltage | VRRM | 100 | V”, “DC Forward Current | IF | 300 | mA”; Vishay, “1N4148”, document 81857 Rev. 1.6, 2024-11-07 (https://www.vishay.com/docs/81857/1n4148.pdf): VRRM 100 V, IF 300 mA, “Forward voltage | IF=10mA | VF | – | – | 1 | V”.

As used: D1 is reversed across U2′s LED and D4 across D3, the bar-live LED. In normal use they see PS2′s output in reverse, a few volts: about 14× under 100 V. They conduct only if PS2 is reversed, through 1 kΩ: a few mA, far under 300 mA.

16.9 The screen, SSD1306 module (U3)

A 0.96″ module printed “GM009605 v4.2”; its maker is unknown, and no maker states the module’s own supply range. For the chip, Solomon Systech, “SSD1306”, Rev. 1.1, April 2008 (https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf):

Table 30. The screen, SSD1306 module (U3).

Item	Value	Page
Logic supply	“V_DD \	Logic Supply Voltage \
Logic inputs	High at 0.8 × VDD or more, low at 0.2 × VDD or less	48, Table 12-1
Absolute maximum supply	“V_DD \	Supply Voltage \
Operating temperature	−40 to +85 °C	47, Table 11-1

As used: U3 runs on `S3_3V3`, at the top of the chip's 1.65 to 3.3 V range and 0.7 V under its 4 V absolute maximum. Its I²C inputs need 2.64 V for a high; the ESP32-S3 guarantees at least 2.64 V out. The two meet exactly on paper.

16.10 The snubber, R1 and C1

As used: at 120 VAC 60 Hz, C1′s 100 nF impedance is about 26.5 kΩ, so about 4.5 mA flows (5.0 mA at 132 VAC, +10 %). R1 dissipates about 2.7 mW (3.2 mW at 132 VAC): about 0.06 % of its 5 W. At switch-on, if the relay closes at the voltage peak with C1 empty, about 1.3 A (1.4 A at 132 VAC) flows for a time constant of 13 µs; the energy in R1 is about 1.4 to 1.7 mJ per closing.

- C1, 100 nF X2, marked 275 VAC (maker unknown). Representative sheet: KEMET, "R46, Class X2, 275 VAC, 110°C", F3093 (https://content.kemet.com/datasheets/KEM_F3093_R46_X2_275_110C.pdf; the edition it serves carries the same texts; the site's certificate may raise a browser warning): "Rated voltage: 275 VAC 50/60 Hz", "X2 CLASS (IEC 60384-14)", "Self healing properties", −40 to +110 °C. Class X2 is tested to 2.5 kV impulses (Vishay, MKP338 4 X2 sheet, document 28118, 2024-07-19, §4.13, https://www.vishay.com/docs/28118/mkp3384x2.pdf). As used: 120 VAC across a 275 VAC part, about 2.3×.
- R1, 130 Ω 5 W, ceramic (cement) body, marked "5W130RJ" (maker unknown). Representative sheet: Yageo, "Cement Resistors, SQP Series / NSP Series", 2021-08-12, in a distributor's copy (https://www.farnell.com/datasheets/3689370.pdf; Yageo's own newer edition, V.5, carries the same quoted texts: https://yageogroup.com/content/datasheet/asset/file/YAGEO-SQP_NSP_DATASHEET), SQP500: "Power Rating at 40 °C | ... 5W", derated linearly to about half at 155 °C; "Maximum Working Voltage | ... 350V"; short-time overload per IEC 60115-1 4.13, "2.5 times RCWV for 5 sec." (about 31 W for 5 s at 130 Ω, about 156 J); "Flameproof ceramic case"; "Excellent pulse load capability." As used: milliwatts steady; each switch-on pulse is about 1/100 000 of that overload test's energy. Rapid repeated switching: not evaluated.

16.11 The bar-sense adapter, JVC AA-R456 (PS2)

- No JVC document for the AA-R456 is known. Its label gives 4.5 V DC 600 mA.
- A resale listing, poweradapter.co, "JVC AA-R451 AC Adapter 4.5V 600mA 5.3W Class 2 Transformer" (the sibling model, not the fitted one, and not the maker), quotes its label: "AC120V~60Hz 5.3W", "DC4.5V === 600mA", "CLASS 2 TRANSFORMER" (https://www.poweradapter.co/jvc-aar451-ac-adapter-45v-600ma-53w-class-2-transformer-p-3468.html).

17 Glossary

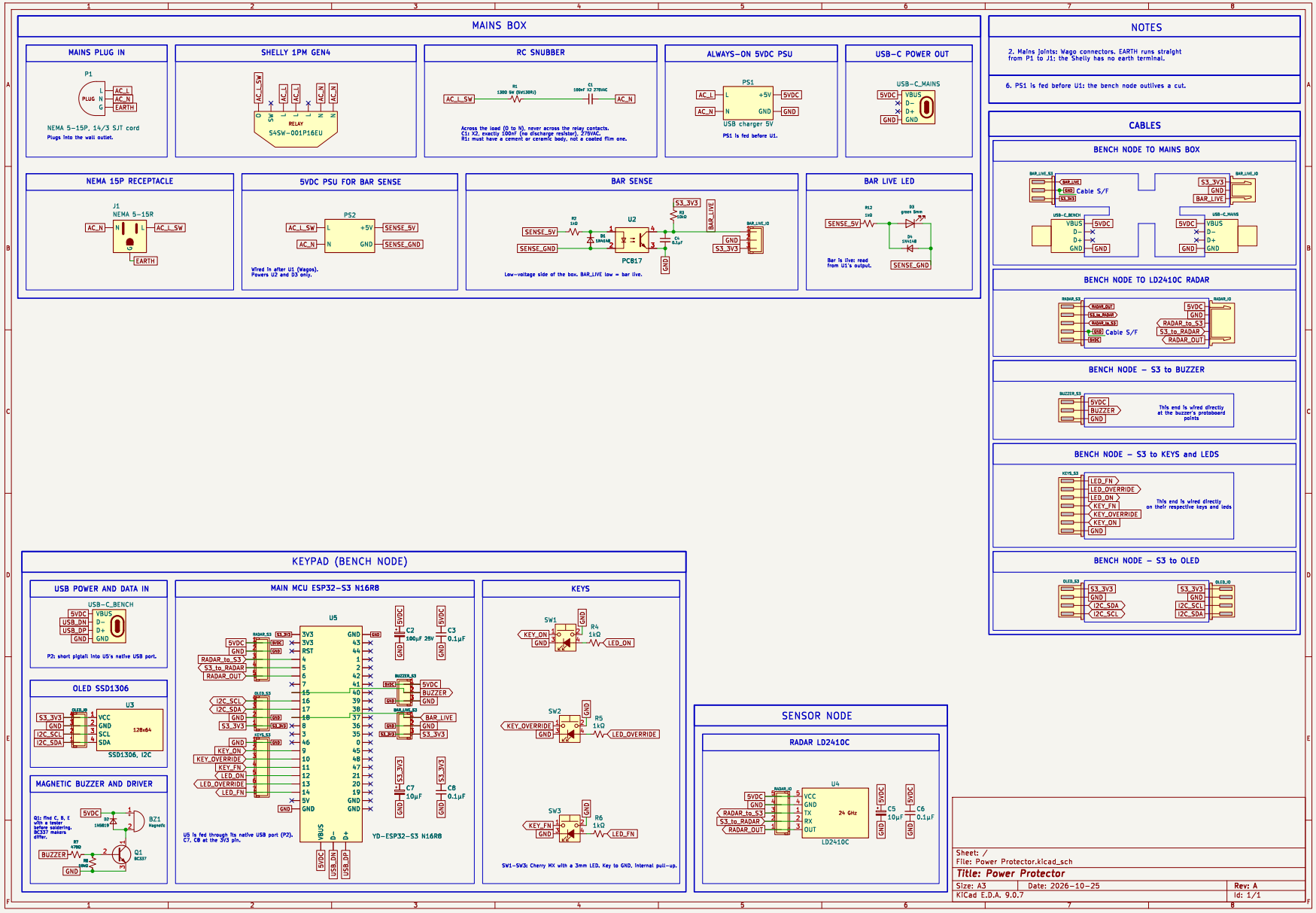
Table 31. Glossary.

Term	Meaning
Bar-live sense	The circuit in the mains box that tells the bench node, independently of the Shelly, whether the bar is live
Bench node	The ESP32-S3 unit at the bench that decides when to cut
Cat7	A category of shielded twisted-pair network cable, with four pairs
Devkit	The ESP32-S3 development board at the heart of the bench node (U5)
HP type	In the relay maker's safety approvals, the high-capacity version of the relay
JST-XH	A small 2.5 mm pitch wire-to-board connector
Lease	The bench node's repeated permission to the Shelly to stay on; when it stops, the Shelly cuts by itself
MOV, varistor	A part that clamps voltage surges
Non-latching relay	A relay that opens as soon as its coil stops being driven
Optocoupler	A part that passes a signal through light, so its two sides share no wire
RC snubber	A resistor and capacitor in series across a load, to tame the spike when an inductive load is switched
RCWV	Rated continuous working voltage of a resistor: the voltage that gives its rated power (the square root of power times resistance), capped at its maximum working voltage
S/FTP	A cable with each pair foiled and an overall braid
Shelly	U1, the Shelly 1PM Gen4 smart relay that switches the bar
SW input	The Shelly's switch input, referenced to mains; not connected in the Deadman Power Protector
TV-5	A UL rating for switching tungsten-filament lamp loads, which draw a large current at switch-on; the number is the steady current class, 5 A
X2 capacitor	A capacitor rated to sit across mains line and neutral

18 The schematic, whole sheet

The whole schematic of hardware revision A follows on its own landscape page (Figure 17).

Figure 17. The whole schematic, revision A.



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5VDC: 2, 3, 5, 6, 9, 12, 15, 17, 20, 21, 22, 24, 26
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AC_L_SW: 9, 11, 19
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BAR_LIVE always reads live (low), even with the bar off: 28
BAR_LIVE reads live long after a cut: 28
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BZ1 is silent: 28
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C6: 6, 17, 20
C7: 5, 12, 18, 20
C8: 5, 12, 20
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The bar is on but BAR_LIVE reads dead at the bench node: 28
The bench node restarts when the buzzer changes pitch: 28
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